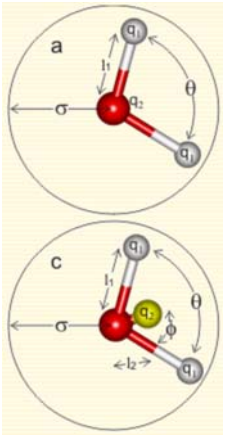
컴퓨터와화학

제5장 분자동역학 모의실험 실습

중요 학습 내용

- VMD 및 NAMD 사용법
- 물분자에 대한 분자동역학 모의실험
- 물상자에서 방사분포함수 구하기
- NaCl 용액에서 방사분포함수 구하기

물분자 모델과 분자동역학 모의실험



paramers	SPC (original)	SPC/E (original)	TIP3P (original)	TIP3P (CHARMM)	TIP4P (original)	TIP4P (2005)
dipole (debye)[a]	2.274	2.351	2.351	2.347	2.18	2.305
$r_{\min}^{OO}$ (Å)	3.5533	3.5533	3.5365	3.5365	3.5399	3.5457
ϵ^{OO} (kcal mol^{-1})	0.1553	0.1553	0.1521	0.1521	0.1549	0.1852
$r_{\min}^{HH}$ (Å)	0	0	0	0.449	0	0
ϵ^{HH} (kcal mol^{-1})	0	0	0	0.046	0	0
$r_{\min}^{OH}$ (Å)	0	0	0	1.993	0	0
ϵ^{OH} (kcal mol^{-1})	0	0	0	0.084	0	0
q^{O} ($\|e\|$)	−0.82	−0.8476	−0.834	−0.834	0	0
q^{H} ($\|e\|$)	0.41	0.4238	0.417	0.417	0.52	0.5564
q^{M} ($\|e\|$)					−1.04	−1.1128
b_0^{OH} (Å)	1.0	1.0	0.9572	0.9572	0.9572	0.9572
b_0^{OM} (Å)					0.15	0.1546
θ_0^{HOH} (deg)	109.47	109.47	104.52	104.52	104.52	104.52
K_b (kcal mol^{-1} Å^{-2})				450.0		
K_θ (kcal mol^{-1} rad^{-2})				55.0		

[a] The experimental value for the isolated water molecule is 1.85 D.

$$V^{H_2O} = \sum_i K_b(r_i - r_{i,0})^2 + \sum_i K_\theta(\theta_i - \theta_{i,0})^2$$
$$+ \sum_{i=1}^{N}\sum_{j>i}^{N} \frac{1}{4\pi\epsilon_0}\frac{q_i q_j}{r_{ij}} + \sum_{i=1}^{N}\sum_{j>i}^{N} \epsilon\left[\left(\frac{r_{\min}^{ij}}{r_{ij}}\right)^{12} - \left(\frac{2r_{\min}^{ij}}{r_{ij}}\right)^{6}\right]$$

NAMD의 input 파일 구성

- PDB 파일: 모의실험 계의 분자들의 좌표를 담고 있는 파일
- PSF 파일: 모의실험 계의 분자들에 관한 좌표 이외의 정보들 (부분전하, 질량 등)을 담고 있는 파일
- CONF 파일: NAMD를 실행하기 위한 환경 (모의실험 상자 크기, 온도, 압력 등)을 정의하는 파일

VMD를 이용해 물 상자 만들기

1) Terminal에서 "vmd"를 입력하여 VMD 띄우기
2) VMD main 창에서, "**Extensions > Modeling > Add Solvation Box**" 선택
3) "Solvate" 창에서 다음을 선택
   - Waterbox Only: Checked
   - Output 이름을 "Solvate" 대신 "watbox"로 입력 (주의: watbox 앞에 파일이 출력될 디렉토리 경로를 모두 기입해야 함)
   - Boundary: 2.4 ← 물분자 사이 최소 거리 2.4 Å 유지
   - Box Size: x, y, z min =(-15.0, -15.0, -15.0) and x, y, z max = (15.0, 15.0, 15.0)
   - Box Padding: 0 Å in all six directions
4) "Solvate" 창 맨 밑의 "**Solvate**" 버튼 클릭 → "watbox.psf"와 "watbox.pdb" 파일 생성 (VMD main 창에 watbox.psf 생성 확인)
5) 생성된 물 상자의 크기와 중심을 확인하기 위해, "**Extensions > Tk Console**" 선택
6) Tk Console 창에서 다음 명령어 입력

   set all [atomselect top all]

   measure minmax $all (상자크기 확인)

   measure center $all weight mass (상자 중심 확인)

Minimization을 위한 Conf 파일 만들기

다음 URL에서 watbox.conf 파일 다운로드https://tinyurl.com/yb4m

Printed by

Nov 06, 17 21:04 **watbox_mini.conf** Page 1/2

```
1  ##############################################################
2  ## JOB DESCRIPTION                                          ##
3  ##############################################################
4
5  # Minimization and Equilibration of
6  # Ubiquitin in a Water Box
7
8
9  ##############################################################
10 ## ADJUSTABLE PARAMETERS                                    ##
11 ##############################################################
12
13 structure          ./watbox.psf
14 coordinates        ./watbox.pdb
15
16 set temperature    0
17 set outputname_pre watbox_mini
18
19 firsttimestep      0
20
21
22 ##############################################################
23 ## SIMULATION PARAMETERS                                    ##
24 ##############################################################
25
26 # Input
27 paraTypeCharmm      on
28 parameters          /opt/local/charmm_ff/charmm27_param/par_all27_prot_lipid.in
   p
29 temperature         $temperature
30
31
32 # Force-Field Parameters
33 exclude             scaled1-4
34 1-4scaling          1.0
35 cutoff              12.0
36 switching           on
37 switchdist          10.0
38 pairlistdist        14.0
39
40
41 # Integrator Parameters
42 timestep            2.0  ;# 2fs/step
43 rigidBonds          all  ;# needed for 2fs steps
44 nonbondedFreq       1
45 fullElectFrequency  2
46 stepspercycle       10
47
48
49 # Constant Temperature Control
50 langevin            on    ;# do langevin dynamics
51 langevinDamping     1     ;# damping coefficient (gamma) of 1/ps
52 langevinTemp        $temperature
53 langevinHydrogen    off    ;# don't couple langevin bath to hydrogens
54
55
56
57
58 # Periodic Boundary Conditions
59 cellBasisVector1    30.0   0.0   0.0
60 cellBasisVector2     0.0  30.0   0.0
61 cellBasisVector3     0.0   0.0  30.0
62 cellOrigin          0.0   0.0   0.0
63
64 wrapAll             on
65
```

Nov 06, 17 21:04 **watbox_mini.conf** Page 2/2

```
66
67 # PME (for full-system periodic electrostatics)
68 PME                 yes
69 PMEGridSpacing      1.0
70
71 #manual grid definition
72 #PMEGridSizeX       45
73 #PMEGridSizeY       45
74 #PMEGridSizeZ       48
75
76
77 # Constant Pressure Control (variable volume)
78 useGroupPressure      yes ;# needed for rigidBonds
79 useFlexibleCell       no
80 useConstantArea       no
81
82 # Output
83 outputName          $outputname_pre.result
84 restartName         $outputname_pre.restart
85
86 binaryoutput         no
87 restartfreq         500     ;# 500steps = every 1ps
88 dcdfreq             250
89 xstFreq             250
90 outputEnergies      100
91 outputPressure      100
92
93
94 ##############################################################
95 ## EXTRA PARAMETERS                                         ##
96 ##############################################################
97
98
99 ##############################################################
100 ## EXECUTION SCRIPT                                        ##
101 ##############################################################
102 # Minimization
103 minimize            1000
```

Monday November 06, 2017 watbox_mini.conf 1/1

How to run NAMD

- Using a single core in a single node

  namd2 conf_file_name > log_file_name &

- Using multi cores in a single node (w/o CUDA)

  namd2 +p4 conf_file_name > log_file_name &

- Using multi cores in a single node with GPU (w/ CUDA)

  namd2 +idlepoll +p4 conf_file_name > log_file_name &

- 예제:

  namd2 +idlepoll +p4 watbox_mini.conf > watbox_mini.log &

MD를 위한 Conf 파일 만들기

다음 URL에서 watbox.conf 파일 다운로드: https://tinyurl.com/yb4mo253

Printed by

Nov 06, 17 21:30 **watbox_md.conf** Page 1/2

```
##############################################################
## JOB DESCRIPTION                                          ##
##############################################################

# Minimization and Equilibration of
# Ubiquitin in a Water Box


#############################################################
## ADJUSTABLE PARAMETERS                                   ##
#############################################################

structure          ./watbox.psf
coordinates        ./watbox_mini.result.coor

set temperature    298.15K
set outputname_pre watbox_md

firsttimestep      0


#############################################################
## SIMULATION PARAMETERS                                   ##
#############################################################

# Input
paraTypeCharmm      on
parameters          /opt/local/charmm_ff/charmm27_param/par_all27_prot_lipid.in
p
temperature         $temperature


# Force-Field Parameters
exclude             scaled1-4
1-4scaling          1.0
cutoff              12.0
switching           on
switchdist          10.0
pairlistdist        14.0


# Integrator Parameters
timestep            2.0  ;# 2fs/step
rigidBonds          all  ;# needed for 2fs steps
nonbondedFreq       1
fullElectFrequency  2
stepspercycle       10


# Constant Temperature Control
langevin            on    ;# do langevin dynamics
langevinDamping     1     ;# damping coefficient (gamma) of 1/ps
langevinTemp        $temperature
langevinHydrogen    off    ;# don't couple langevin bath to hydrogens




# Periodic Boundary Conditions
cellBasisVector1    30.0   0.0   0.0
cellBasisVector2     0.0  30.0   0.0
cellBasisVector3     0.0   0.0  30.0
cellOrigin           0.0   0.0   0.0

wrapAll             on
```

Nov 06, 17 21:30 **watbox_md.conf** Page 2/2

```
# PME (for full-system periodic electrostatics)
PME                 yes
PMEGridSpacing      1.0

#manual grid definition
#PMEGridSizeX        45
#PMEGridSizeY        45
#PMEGridSizeZ        48


# Constant Pressure Control (variable volume)
useGroupPressure      yes ;# needed for rigidBonds
useFlexibleCell       no
useConstantArea       no

# Output
outputName          $outputname_pre.result
restartName         $outputname_pre.restart

binaryoutput         no
restartfreq         500     ;# 500steps = every 1ps
dcdfreq             250
xstFreq             250
outputEnergies      100
outputPressure      100

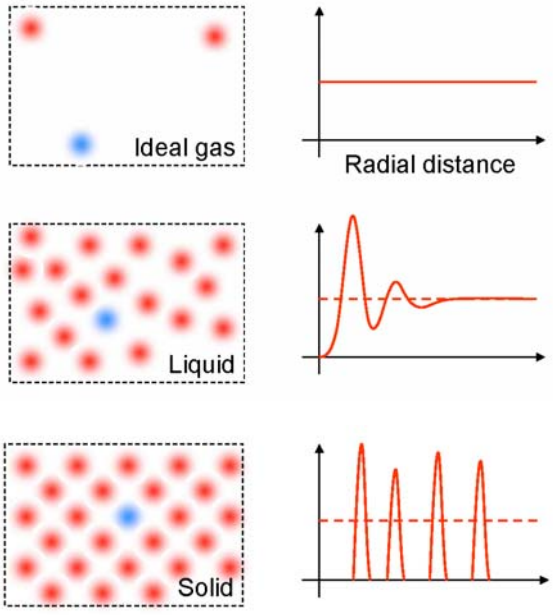

#############################################################
## EXTRA PARAMETERS                                        ##
#############################################################

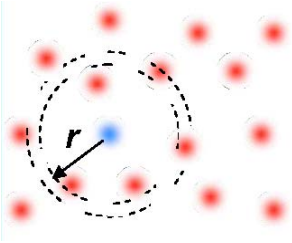
#############################################################
## EXECUTION SCRIPT                                        ##
#############################################################
# MD run
numsteps            100000
```

Monday November 06, 2017 watbox_md.conf 1/1

방사분포함수 (Radial distribution function, RDF) - structured fluids

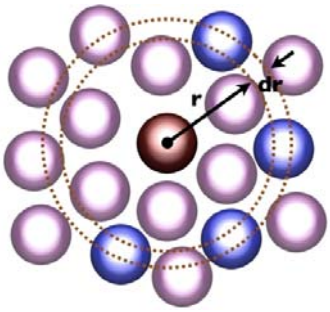
- Radial distribution function of gas, liquid, and solid

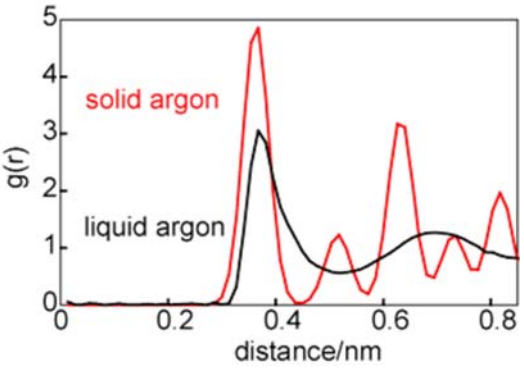


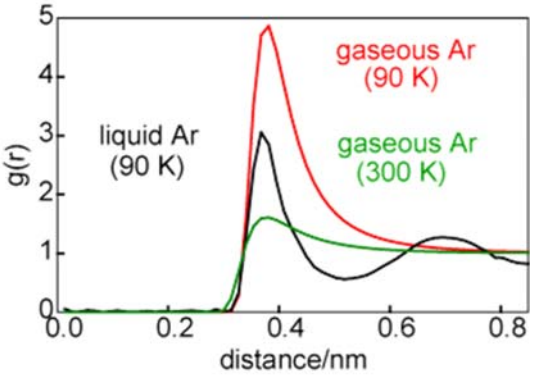


- Calculate the number of particles inside a radial segment ***(r,r+Δr)*** from one specific particle in the system.
- Average the distribution over all particles in the system.
- Divide the (average) number of particles in a segment by the segment volume.
- Finally, normalize thus obtained radial density profile with the bulk density ***N/V*** of the system.

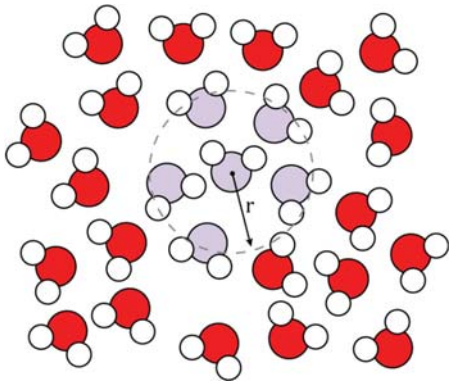
RDF of argon

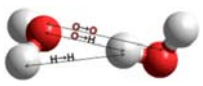


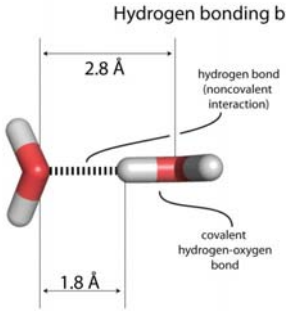


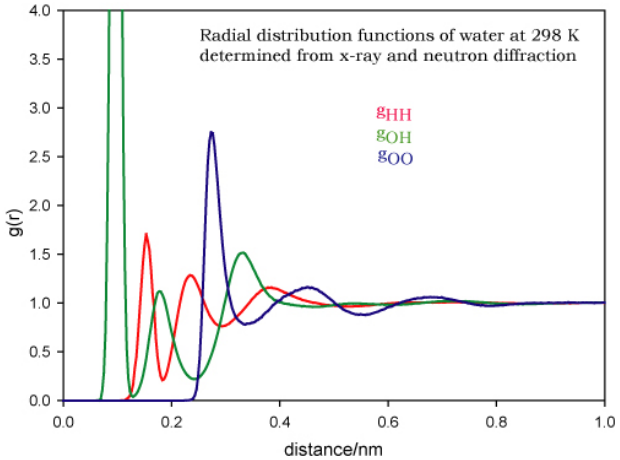


중성자 산란 실험으로부터 얻어진 물의 **RDF**

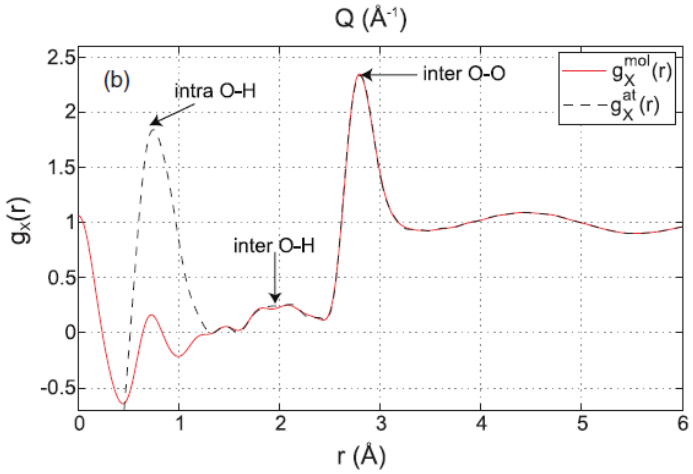








X선 회절실험으로 구한 물의 RDF



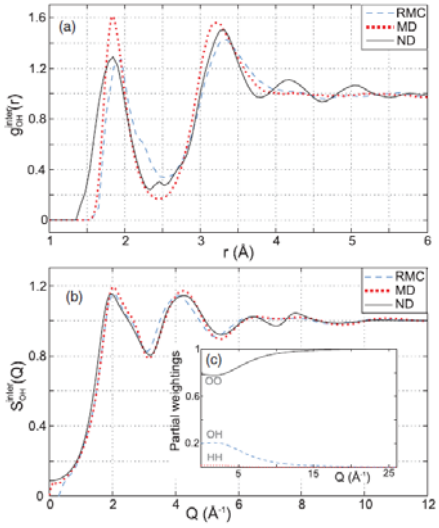
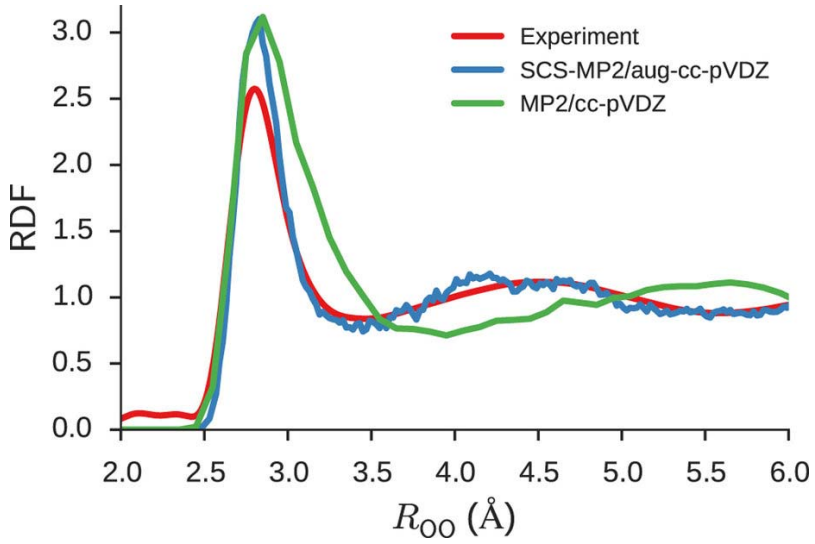


FIG. 7. Comparison of the different $g_{OH}(r)$ (a) and the corresponding $S_{OH}(Q)$ (b) from the literature. The light blue dashed line is the result from RMC fitting in the present work to both the *APS* dataset and neutron diffraction data from Ref. 12, the red dotted line is the result from the TIP4P/2005 MD simulation,[36] and the black solid line is the result from recent neutron diffraction data with oxygen isotope substitution.[33,34] Note that the intramolecular O-H contribution is not included in this figure. A $Q_{max} = 24$ Å^{-1} was used in the Fourier transforms to derive $g_{OH}(r)$ from the neutron diffraction data.[33] As a consequence, the O-H peak around 2 Å is artificially broadened in the neutron $g_{OH}(r)$ measurement, whereas the RMC and TIP4P/2005 MD $g_{OH}(r)$ functions are not obtained by the Fourier transform and hence are not artificially broadened. (c) Inset: The Q-dependent partial weighting factors of the O-O correlations (black solid line), the O-H correlations (blue dashed line), and the H-H correlations (red dotted line).

J. Chem. Phys. **138**, 074506 (2013).

Ab initio MD로부터 구한 물의 RDF



Sci. Rep. **5**, 14358 (2015).

VMD를 이용해 H_2O의 RDF 구하기

1) VMD main 창에서, "**Extensions > Analysis > Radial Pair Distribution Function g(r)**" 선택

2) "Radial Pair Distribution Function g(r)" 창에서 다음을 선택
   - Use Molecule에서 "**0:watbox.psf**" 선택
   - $g_{O\text{-}O}(r)$ 그래프 그리기: Selection 1에 "**type OT**", Selection 2에 "**type OT**" 기입
   - $g_{O\text{-}H}(r)$ 그래프 그리기: Selection 1에 "**type OT**", Selection 2 에 "**type HT**" 기입
   - $g_{H\text{-}H}(r)$ 그래프 그리기: Selection 1에 "**type OT**", Selection 2에 "**type HT**" 기입
   - Frames: "First:" 에 "100" 입력

3) "Radial Pair Distribution Function g(r)" 창 맨 밑의 "**Compute g(r)**" 클릭

VMD를 이용해 NaCl 용액 만들기

1) Terminal에서 "vmd"를 입력하여 VMD 띄우기
2) VMD main 창에서, "**Extensions > Modeling > Add Ions**" 선택
3) "Solvate" 창에서 다음을 선택
   - Input PSF와 PDB: 각각 "watbox.psf"와 "watbox.pdb" 선택
   - Output prefix: "Ionized" 대신 "nacl"로 입력 (주의: nacl 앞에 파일이 출력될 디렉토리 경로를 모두 기입해야 함)
   - Choose salt: "NaCl"
   - Ion placement mode: "**Neutralize and set ion concentration to**" 선택, 농도 "0.5 mol/L" 선택
4) "Add Ions" 창 맨 밑의 "**Autoionize**" 버튼 클릭 → "nacl.psf"와 "nacl.pdb" 파일 생성 (VMD main 창에 nacl.psf 생성 확인)

VMD를 이용해 NaCl의 RDF 구하기

1) VMD main 창에서, "**Extensions > Analysis > Radial Pair Distribution Function g(r)**" 선택

2) "Radial Pair Distribution Function g(r)" 창에서 다음을 선택
   - Use Molecule에서 "0:nacl.psf" 선택
   - $g_{na\text{-}O}(r)$ 그래프 그리기: Selection 1에 "name SOD", Selection 2에 "type OT" 기입
   - $g_{na\text{-}H}(r)$ 그래프 그리기: Selection 1에 "name SOD", Selection 2 에 "type HT" 기입
   - $g_{cl\text{-}O}(r)$ 그래프 그리기: Selection 1에 "name CLA", Selection 2에 "type OT" 기입
   - $g_{cl\text{-}H}(r)$ 그래프 그리기: Selection 1에 "name CLA", Selection 2 에 "type HT" 기입
   - $g_{na\text{-}cl}(r)$ 그래프 그리기: Selection 1에 "name SOD", Selection 2에 "name CLA" 기입
   - $g_{na\text{-}na}(r)$ 그래프 그리기: Selection 1에 "name SOD", Selection 2 에 "name CLA" 기입
   - $g_{cl\text{-}cl}(r)$ 그래프 그리기: Selection 1에 "name SOD", Selection 2에 "name CLA" 기입

3) "Radial Pair Distribution Function g(r)" 창 맨 밑의 "Compute g(r)" 클릭

Na^+, Cl^- 이온과 물분자와의 g(r)

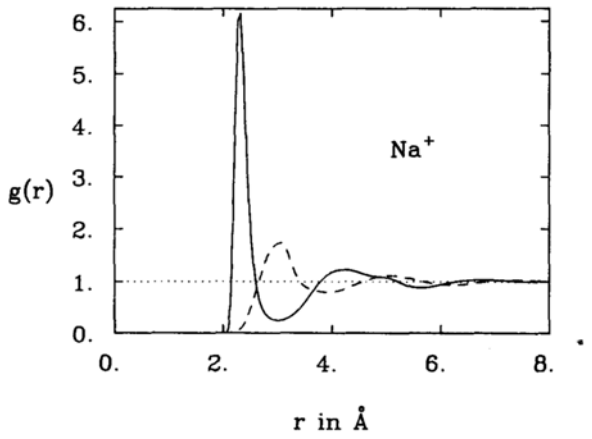


FIG. 1. Radial solute–solvent pair distribution functions for Na^+–H_2O. The solid line is $g_{NaO}(r)$. The dashed line is $g_{NaH}(r)$.

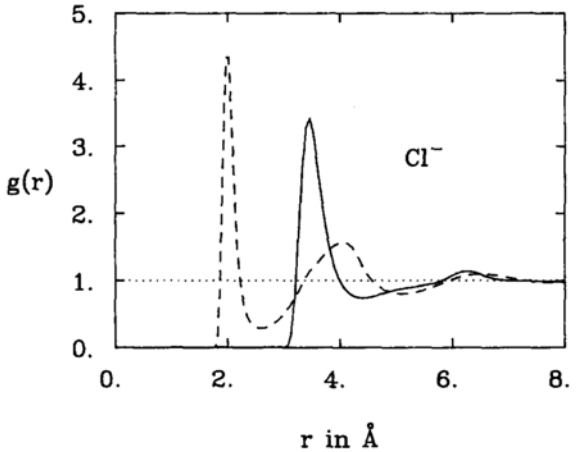
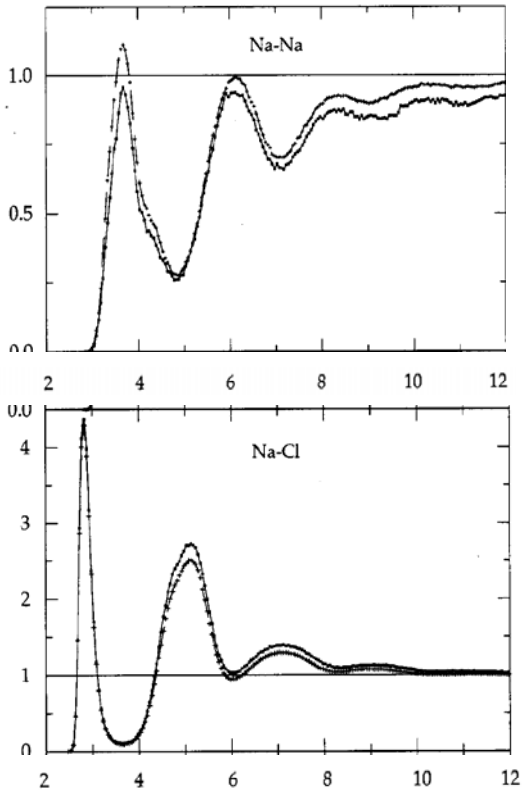


FIG. 2. Radial solute–solvent pair distribution functions for Cl^-–H_2O. The solid line is $g_{ClO}(r)$. The dashed line is $g_{ClH}(r)$.

Pettitt & Rossky, J. Chem. Phys. **84**, 5836 (2016).

Na^+-Na^+, Cl^--Cl^-, Na^+-Cl^- 의 g(r)



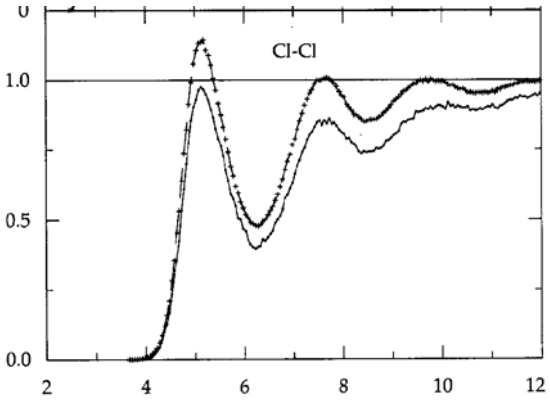


FIG. 1. Radial distribution functions for ion pairs in aqueous NaCl electrolyte. Continuous line, 0.5 M; dashed line, 1 M.

Lyubartsev & Marcelja, Phys. Rev. E, **65**, 041202 (2016).